Dr. Pushpendra Singh Sisodia

Principal Data Scientist

Ahmedabad, India | +91-7500065050

https://pushpendra-muj.github.io/myprofile.github.io/

Pushpendra.muj@gmail.com | www.linkedin.com/in/dr-pushpendra-sisodia

Profile / Summary Statement

Principal Data Scientist with **19 years of overall experience** and **12 years specializing in AI/ML**, delivering enterprise scale machine learning and **Generative AI solutions** from research to production. Proven expertise in leading end-to-end data science programs including **RAG-based knowledge systems, agentic workflow automation, LLM evaluation/governance frameworks**, and classical ML across diverse business domains. Strong hands-on capability in designing and deploying production-grade AI using **Python, PyTorch, TensorFlow**, and **AWS (SageMaker, Bedrock, Lambda, ECS/EKS, Redshift, S3)** with a focus on scalability, reliability, and measurable business outcomes. Experienced in **LLMOps/MLOps practices** including CI/CD, automated evaluation, monitoring, drift detection, model/version management, and governance for safe enterprise adoption. Adept at translating stakeholder needs into technical solutions, mentoring teams, and driving AI strategy through cross-functional collaboration. Passionate about building trustworthy AI systems that automate decision-making, improve operational efficiency, and accelerate organizational AI maturity.

Key Skills & Areas of Expertise

Generative AI & LLM	RAG, LLMs, Hybrid Search (Dense + Keyword), Embedding, Chunking Strategy, Metadata Tagging, Re-ranking, Summarization, Classification, Prompt Engineering, Prompt Versioning
Agentic AI	Tool-Using Agents, Multi-Step Workflows, Planner–Executor Orchestration, Tool Invocation, Workflow Automation, Structured Output Validation (JSON Schema), Retry/Fallback Strategies
LLMOps & Governance	LLM Evaluation (Groundedness, Faithfulness, Hallucination Detection), Citation Quality Scoring, Safety Evaluation (Toxicity, Bias, Guardrails), Regression Testing, Governance Frameworks, Audit Logs, Monitoring & Observability
ML & Applied AI	Supervised/Unsupervised Learning, NLP, Computer Vision, Time Series Forecasting, Anomaly Detection, Numerical Optimization, Feature Engineering, Model Selection, Experimentation
AWS Cloud	Amazon SageMaker, Bedrock, Lambda, Step Functions, ECS, EKS, S3, OpenSearch, Redshift, CloudWatch, Event-Driven Architectures
MLOps & Production	CI/CD, Automated Testing, Model Deployment, Model Versioning, Drift Detection, Retraining Pipelines, Experiment Tracking, Reproducibility, Containerization
Programming & Tools	Python, SQL, PyTorch, TensorFlow, Docker, Kubernetes, Git
Leadership	End-to-End Project Leadership, Stakeholder Management, Cross-functional Collaboration, Mentoring & Coaching, AI Strategy & Roadmap Planning, Executive Communication

Professional Experience

Topia Life Sciences Pvt. Ltd. *Ahmedabad, India*

Principal Data Scientist *Jun. 2023 – Present*

Project-1 (Drug Repurpose) Enterprise RAG Copilot (Production-Grade Knowledge Assistant) (*LLM*)

- **Amazon Bedrock + OpenSearch + S3: production-grade RAG copilot to** answer enterprise queries from PDFs, SOPs, Confluence, and structured sources
- Designed **hybrid retrieval** (dense embeddings + keyword search + reranking) and optimised chunking/metadata strategies to improve answer grounding and **reduce hallucinations**.
- Built automated ingestion pipelines using **S3 + AWS Lambda**, enabling scalable document processing, embedding generation, indexing, and version-controlled updates.
- Implemented **LLM evaluation + safety checks** (faithfulness, groundedness, citation quality, toxicity) with regression tests for prompt/model updates.
- Optimised performance by reducing latency via caching, batch embedding generation, and retrieval tuning; deployed on **ECS/EKS** with monitoring in **CloudWatch** and analytics in **Redshift**.

Project-2 (Alzevita) Hippocampus Segmentation & Volumetric Analysis Software (FDA 510(k) Cleared) (*Deep Learning*)

- Lead development of a **deep-learning-based hippocampus segmentation and volumetric analysis** software, from architecture design through regulatory clearance to market release.
- Design and train **3D U-Net** models for automated volumetric segmentation of brain MRI scans, optimising for accuracy, reproducibility, and clinical usability.
- Drive technical validation and documentation supporting the product's **FDA 510(k) clearance**, coordinating with regulatory and clinical teams on performance testing and evidence generation.
- Currently leading development of a **whole-brain segmentation** module, extending the platform's regional segmentation capability to full-brain structural analysis.
- Build and validate volumetric analysis pipelines that translate segmentation outputs into clinically interpretable metrics for neurologists and researchers.

Project- 3 (SMAG) NCE Discovery Platform: AI/ML Toolkit for Novel Chemical Entity Identification (*Machine Learning*)

- Design and develop an integrated suite of **machine learning tools** that help identify and prioritise Novel Chemical Entities (NCEs), spanning the drug discovery pipeline from target screening to lead optimisation.
- Build **QSAR (Quantitative Structure Activity Relationship)** models to predict compound bioactivity and guide structure-based optimization decisions.
- Develop **molecular docking** and **ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) prediction models** to filter and rank candidate compounds early in the discovery pipeline.
- Implement **Multi-Parameter Optimisation (MPO)** frameworks to balance potency, selectivity, and drug-like properties across candidate molecules.
- Build **retrosynthesis prediction and scaffold-hopping models** to support synthetic route planning and generation of novel chemical scaffolds.

Indus University *Ahmedabad, India*

Associate Professor *Jul. 2021 – May 2023*

Project – LLM Evaluation + Governance Framework (LLMOps Quality & Safety Platform)

- Built a reusable **LLM evaluation + governance framework** to standardise quality, safety, and reliability across RAG, summarisation, classification, and insight-generation systems.
- Implemented automated evaluation for **hallucination detection**, groundedness, citation correctness, toxicity checks, and response consistency across model versions.
- Developed **regression testing pipelines** for prompt changes, retrieval updates, and LLM upgrades to prevent performance degradation in production.
- Implemented monitoring dashboards for latency, failure rates, retrieval drift, user feedback, and business KPIs using **CloudWatch + Redshift**.
- Optimised evaluation performance by batching test runs, parallelising scoring pipelines, and enabling lightweight smoke tests for **CI/CD release gates**.

Poornima College of Engineering *Jaipur, India*

Associate Professor *2019 – 2021*

Project – Agentic Workflow Automation System (Tool-Using Multi-Step AI Agents)

- Designed and deployed an **agentic AI automation system** using **tool-using agents** to execute multi-step workflows across structured and unstructured enterprise data.
- Implemented orchestration patterns (planner–executor, tool invocation, workflow retries) using **AWS Step Functions + Lambda** for scalable event-driven execution.
- Integrated enterprise APIs and data sources (databases, dashboards, document stores) with schema-driven outputs (**JSON validation**) to ensure reliability and traceability.
- Built **evaluation pipelines** measuring task success rate, tool accuracy, workflow latency, and failure-mode analysis to improve production stability.
- Optimized runtime performance by minimizing tool calls, adding intermediate caching, and implementing **timeout/fallback strategies** for safe automation.

Associate Professor – Anand Int. College of Engineering, Jaipur (2016–2019)
Associate Professor – Poornima University, Jaipur (2014–2015)
Teaching Associate – Manipal University Jaipur (2012–2014)
Assistant Professor – KITE, Jaipur (2011–2012)
Assistant Professor – SGVU, Jaipur (2008–2011)
Web Developer – Visual Solutions, Jaipur (2007–2008)

Education

Postdoc in Artificial Intelligence – National Taipei University of Business *2022–2023*
PhD in Computer Science – Manipal University Jaipur *2012–2018*
M.Tech. in Software Engineering – SGVU, Jaipur *2009–2011*
B.E. in Information Technology – Rajasthan University *2003–2007*

Selected Publications

Performance of machine learning algorithms for lung cancer prediction: a comparative approach	*Scientific Reports (Nature)*	Sisodia P., et al. 2024
A review of deep transfer learning approaches for class-wise prediction of Alzheimer's disease using MRI images	*Archives of Computational Methods in Engineering (Springer Nature)*	Sisodia P., et al. 2023
A comprehensive analysis of artificial intelligence techniques for the prediction and prognosis of genetic disorders using various gene disorders	*Archives of Computational Methods in Engineering (Springer Nature)*	Sisodia P., et al. 2023
A deep transfer learning based approaches for the detection and classification of acute lymphocytic leukemia using microscopic images	*Multimedia Tools and Applications (Springer)*	Sisodia P., et al. 2022

Languages

English (Fluent), Hindi (Native)

References

References available upon request.